

CLASSROOM MANAGEMENT AND BROWSING SYSTEM

Current Classroom Issues

Main daily issues students and faculty face when looking for classrooms

- Searching Room by Room

Students and instructors often walk through hallways checking doors, which wastes valuable time between lectures.

- Unused Classrooms

Many rooms stay empty for hours simply because people do not know they are available to use.

- Unknown Room Capacity

Users cannot check the number of seats before arriving, making it hard to choose a room that fits their group size.

Review of Existing Systems

An overview of current campus scheduling and room management solutions

-University Academic Scheduling (e.g., Banner, Coursedog):

Designed primarily for university management for semester scheduling

Highly complex and lacks a quick lookup interface for students and faculty.

-Facility & Room Reservation Tools (e.g., LibCal, Resource Central):

Focuses on booking workflows and reservation forms for labs and study rooms.

Requires prior booking steps and approvals rather than fast, instant browsing.

-Manual & Static Solutions (e.g., Excel sheets, printed door schedules):

Widely used on campuses, but static, prone to errors, and often outdated



Our Proposed Solution

How our system solves these classroom issues in 4 simple steps

- Simple Browsing Flow

Users select a building, see all its rooms, and check free hours without any complex steps.

- Clear Capacity Information

Shows the exact number of seats in advance to help users choose a room that fits their group size.

- Visible Daily Schedules

Displays occupied and free hours clearly so empty classrooms can be found and used easily

- Fast and Easy Access

Eliminates the need for prior approvals or formal booking steps, allowing immediate use of available rooms.

Feature Comparison

FEATURE	UNIVERSITY SYSTEMS (E.G., BANNER)	BOOKING TOOLS (E.G., LIBCAL)	OUR PROPOSED SYSTEM
ROOM SELECTION BY CAPACITY	Stores capacity as data, but does not help users choose a room based on group	Displays capacity but lacks filtering by group size.	Helps users easily pick a suitable room based on the number of people.
CHECKING AVAILABLE TIMES	Complex schedules that are hard to read for daily use.	Requires opening a calendar and starting a booking process	Displays occupied and available hours clearly in one simple view
BROWSING BY BUILDING	Displays all campus rooms in long, complicated lists.	Often organized by room type rather than by building location	Lets users select a building first, then view all its classrooms.
MAIN PURPOSE	Designed for administrative planning and semester schedule	Built for formal reservations, permissions, and study room booking	A quick, lightweight browsing tool for students and faculty

User Personas



Students

Description:

University students who need a place to study or hold group meetings.

Example:

Sara, a software engineering student, needs a room for a meeting with 5 classmates. She selects a building, sees the rooms that fit 6 people, and checks which ones are free at 10:00.



Faculty Members

Description:

Instructors who need suitable rooms for lectures, activities, and meetings.

Example:

Dr. Ahmed needs a room for a makeup lecture with 40 students. He finds that Room 101 has a capacity of 40 and is available from 10:00 to 12:00.

User Personas



University Administration

Description: Officials who monitor how classrooms are used and organized.

Example: Mr. Khalid wants to know if large rooms are actually being used. He reviews occupied and free times per building and notices many rooms are empty in the afternoon.



Room Management and Scheduling Departments

Description: The units responsible for managing room information and schedules.

Example: Eng. Noura adds a new room in Building 10 with a capacity of 30 students, then updates its occupied times after the semester schedule changes.

Functional Requirements

- Allow users to select the building where the classroom is located.
- Allow users to specify the number of students who will use the classroom.
- Allow users to enter the date and time for the classroom reservation.
- Show users a list of all available classrooms in the selected building for the specified date and time, considering the required capacity and the university's operating hours.
- Allow users to reserve an available classroom in the selected building for the specified date and time.

Non-Functional Requirements

1. Usability:

The interface is simple and clear, making it easy for users to interact with the system.

2. Response Time:

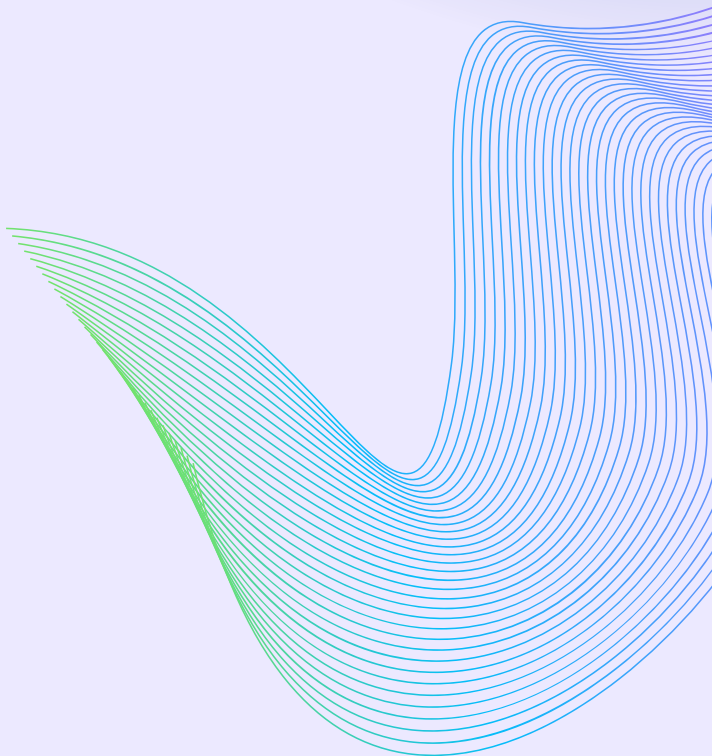
The system displays search results quickly.

3. Availability:

The system can be accessed whenever it is needed.

4. Maintainability:

The classroom and schedule data can be easily updated when needed.



System Workflow:

1 Open the App

Launch the app and tap Book a Room.



2 Select a Building

Choose the building where you want to book.



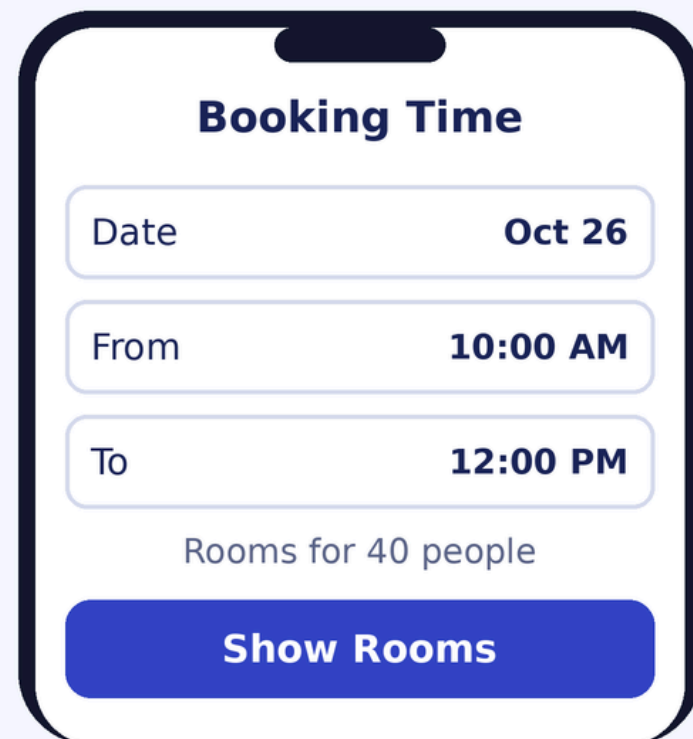
3 Number of People

Enter the group size to see rooms that fit.



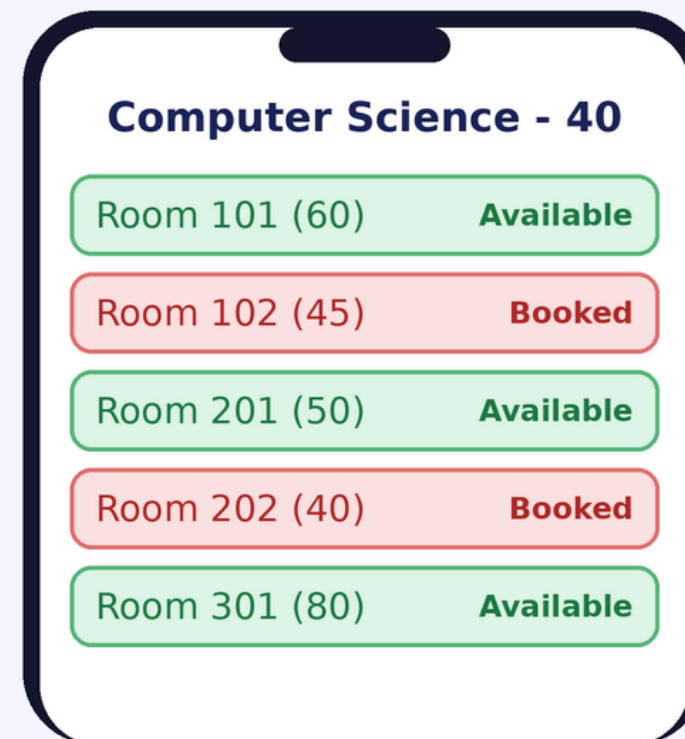
4 Choose the Time

Pick the date, start time and end time.



5 Available Rooms

See suitable rooms: available or booked.



6 Room Time Slots

Check all time slots, then book the room.





THANK
YOU